

DevelopersHub Corporation

Data Science & Analytics Internship Tasks

Due Date: 15th May, 2026

Overview

As part of your Data Science & Analytics Internship at DevelopersHub Corporation, you are required to complete **at least 3 out of the 5 tasks** listed below. You are encouraged to complete all of them if you are interested in gaining more hands-on experience.

These tasks are designed to help you develop core skills in data science, such as data exploration, visualization, model building, and performance evaluation using Python. You will use libraries like `pandas`, `matplotlib`, `seaborn`, and `scikit-learn`.

Task 1: Exploring and Visualizing a Simple Dataset

Objective:

Understand how to read, summarize, and visualize a dataset.

Dataset:

Iris Dataset (CSV format, available through seaborn or other open sources)

Instructions:

- Load the dataset using the `pandas` library.
- Display dataset structure using `.shape`, `.columns`, and `.head()`.
- Create basic visualizations:
 - Scatter plot to analyze relationships between variables.
 - Histogram to examine data distribution.
 - Box plot to detect outliers and spread of values.

- Use `matplotlib` and `seaborn` for visualizations.

Skills:

- Data loading and inspection using pandas
- Basic data summarization
- Visualization using matplotlib and seaborn

Task 2: Credit Risk Prediction

Objective:

Predict whether a loan applicant is likely to default on a loan.

Dataset:

Loan Prediction Dataset (available on Kaggle)

Instructions:

- Handle missing data appropriately.
- Visualize key features such as loan amount, education, and income.
- Train a classification model like Logistic Regression or Decision Tree.
- Evaluate the model using accuracy and a confusion matrix.

Skills:

- Data cleaning and handling missing values
- Exploratory Data Analysis (EDA)
- Binary classification using machine learning
- Model evaluation using confusion matrix and accuracy

Task 3: Customer Churn Prediction (Bank Customers)

Objective:

Identify customers who are likely to leave the bank.

Dataset:

Churn Modelling Dataset

Instructions:

- Clean and prepare the dataset.
- Encode categorical features such as geography and gender.
- Train a classification model.
- Analyze feature importance to understand what influences churn.

Skills:

- Categorical data encoding (Label Encoding / One-Hot Encoding)
- Supervised classification modeling
- Understanding and interpreting feature importance

Task 4: Predicting Insurance Claim Amounts

Objective:

Estimate the medical insurance claim amount based on personal data.

Dataset:

Medical Cost Personal Dataset

Instructions:

- Train a Linear Regression model to predict charges.
 - Visualize how BMI, age, and smoking status impact insurance charges.
 - Evaluate model performance using MAE and RMSE.
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Skills:

- Regression modeling
- Feature correlation and visualization
- Error evaluation using MAE and RMSE

Task 5: Personal Loan Acceptance Prediction

Objective:

Predict which customers are likely to accept a personal loan offer.

Dataset:

Bank Marketing Dataset (UCI Machine Learning Repository)

Instructions:

- Perform basic data exploration on features such as age, job, and marital status.
- Train a Logistic Regression or Decision Tree classifier.
- Analyze the results to identify which customer groups are more likely to accept the offer.

Skills:

- Data exploration and visualization
- Classification modeling
- Business insight extraction from data

Submission Requirements (Checklist for Each Task)

To receive credit for each completed task, ensure the following items are included:

1. Jupyter Notebook

- Introduction and problem statement
- Dataset understanding and description
- Data cleaning and preparation
- Exploratory Data Analysis (EDA) with graphs
- Model training and testing
- Evaluation metrics (e.g., accuracy, confusion matrix, MAE, RMSE, etc.)
- Conclusion summarizing key insights

2. Code Quality

- Code should be clean, well-structured, and include comments explaining each step.

3. GitHub Repository

- Create a dedicated GitHub repository for your internship tasks.
- Give the repository a clear and descriptive name.
- Add a [README.md](#) file summarizing:
 - The task objective
 - Your approach
 - Results and insights

4. Submission on Google Classroom

- Submit the link to your GitHub repository for each completed task.

Important Note

- **Complete at least 3 out of 5 tasks** before the due date: **15th May, 2026**

- You may complete all 5 tasks for more practice and stronger portfolio.

- Ask for help whenever needed. Mentors are here to guide you through.